Robed Beauvile-Owen, Ph.D.

Data Scientist | Measurement, Experimentation & Predictive Modeling

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EXPERIENCE

Founder & Data Scientist — *AniPoint Site Research* Pittsburgh, PA | May 2026 - Present

- Founded a location intelligence consultancy that measures demand and opportunity across counties to support site selection, with feasibility screenings that steered clients away from low-potential sites.
- Translated owners' profit goals into required customer-volume targets with a revenue driver tree (traffic × capture rate × average ticket), and quantified each site's downside risk with Monte Carlo simulation (probability of loss, VaR, CVaR).
- Estimated that each new location adds $533K in market revenue a year, using Double ML (econml, XGBoost) validated with difference-in-differences and a synthetic-data recovery test.
- Designed a site-scoring model that ranks locations on demographic, traffic, competitive-density, and consumer-behavior features, reported to owners with explicit assumptions and uncertainty ranges in written reports and maps.
- Engineered scalable, reusable Python ETL pipelines that process Census, traffic, and competitor data for each new market, with automated data-quality checks and 250+ unit tests.

Senior Data Scientist — *Limetree* Dallas, TX (Remote) | September 2024 - April 2026

- Built an in-house analysis platform that measured experiment performance to guide campaign budgets, automating power analysis, significance tests, and A/B test readouts, with CUPED narrowing confidence intervals (20% lower variance).
- Built automated reporting tools for monthly and weekly reporting, metric monitoring, and Power BI dashboards used by data science and strategy.
- Standardized core KPIs (gross lead rate, enrollment rate, cost per member), fixing conflicting lead and attribution rules with data engineering.
- Improved campaign targeting with uplift and segmentation models on millions of HIPAA-protected (PII/PHI) Medicare/Medicaid-eligible records, adopted in 90% of campaigns.
- Re-architected the production churn prediction pipeline in Python, SQL, and Spark on Databricks, cutting inference runtime 48% and memory 23%.

Founding Machine Learning Engineer — *eParts Services LLC* Pittsburgh, PA | September 2023 - September 2024

- As the first ML hire, built predictive pricing models that improved shipping estimates, cutting median error vs. carrier cost from $13 to $4 on standard orders while maintaining contribution margin.
- Deployed a regression pricing API in C# on Azure (10,000+ monthly transactions), trained on SQL tables that split bulk orders into item-level costs.
- Tuned Elasticsearch relevance (custom analyzers, field boosting) to cut search dissatisfaction 12% and raise click-through rate, validated with NDCG@10 and A/B tests.

Data Scientist — *Limetree* Dallas, TX (Remote) | April 2022 - September 2023

- Built XGBoost lead-scoring models on millions of healthcare consumer records with hundreds of external attributes, raising offline AUC-ROC, explaining drivers with SHAP, and confirming a 9% GLR lift in an A/B test.
- Automated reporting pipelines and Power BI dashboards with Python, SQL, and Snowflake, plus a company-wide self-service dashboard.
- Owned the monthly model monitoring report (drift, accuracy, forecast vs. actual) for up to 5 production models, read by the managing partner.
- Doubled the precision of high-opportunity zone selection (15% to 30%, measured against past successful campaigns) with GeoPandas spatial analysis.

TECHNICAL PROJECTS

Medicare Claims Intelligence github.com/beauvilerobed/.../projects/medicare-claims-intelligence

- Built a dbt warehouse with 26 data tests over 790K synthetic CMS Medicare claims and a fraud (FWA) detector checked against planted billing schemes.
- Forecast weekly flu-like illness visits from CDC data with 41% lower error than a seasonal baseline in a rolling backtest.
- Built an explainable XGBoost model (SHAP) that finds 46% more next-year high-cost members than ranking by past cost.

Customer LTV & Churn Survival Modeling github.com/beauvilerobed/.../projects/ltv-churn-survival

- Showed why the standard LTV = ARPU / churn formula misleads by simulating 1M customer lifetimes in Python, where it held under constant churn but failed once churn dropped with customer tenure, and replaced it with lifetime value estimated from empirical survival curves.

CaLaun — *AI-Powered Calculus Tutor (LLM Application)* github.com/calaun-learning/calaun

- Built a generative AI tutor that grounds a self-hosted Llama 3.2 LLM in SymPy-verified solution steps, so the model explains but never computes.

EDUCATION

Ph.D. in Mathematics — *University of Pittsburgh* 2017 - Dec 2023

Dissertation: A Minimal Invariant Set, Frames, and Operators

B.S. in Mathematics — *Florida State University* 2014 - 2017

Minor: Philosophy | **GPA:** 3.90

TECHNICAL SKILLS

Measurement & Metrics: KPI Frameworks, Driver Trees, Metric Definitions, Leading Indicators, Time Series Forecasting (ARIMA, ETS, Prophet), Monte Carlo Simulation, Risk Metrics (VaR, CVaR), Model Monitoring & Drift

Experimentation & Causal Inference: A/B Testing, Experimentation Platforms, Power Analysis, CUPED, Hypothesis Testing, Uplift Modeling, Double ML, Causal Forests, Difference-in-Differences, CATE

Predictive Modeling & AI: Regression, Classification, Segmentation, XGBoost, Survival Analysis, Anomaly Detection, SHAP, Feature Engineering, Optimization, NLP, LLMs (Llama), Prompt Engineering

Programming & Data: Python (pandas, NumPy, scikit-learn, statsmodels, econml, GeoPandas), SQL, C#, Bash, PowerShell, Spark, Databricks (Workflows), Snowflake, dbt, MLflow, Azure, SQL Server, PostgreSQL, Docker, Elasticsearch, ETL/ELT, Data Quality Checks, Unit Testing, Git, Claude Code

Visualization & Communication: Power BI, Tableau, Excel, Matplotlib, Plotly, Dashboards, Stakeholder Presentations, Mentoring